



CONTEXT & OBJECTIVES



Figure 1.1 Inverter in PV Farm

100k
PV Systems

#1
Reported Failures Inverter

- Inverters were the most frequently reported failing component across 100,000 PV systems [1]
- Thermal issues are a frequent source of inverter failure [2]
- Existing monitoring methods are largely reactive and lack spatial monitoring.

1. Physics Based Digital Twin → Simulate electrothermal behaviour under conditions
2. Raman Distributed Temperature Sensing → Spatial thermal fault localisation
3. Predictive Maintenance Framework → Combining methods to predict degradation

HYPOTHESIS & METHODOLOGY

Hypothesis: Combining these methods will enable earlier and more reliable detection of inverter thermal degradation than conventional temperature monitoring alone.

Electrothermal Digital Twin

- Three Phase String Inverter Model made on MATLAB and Visualised
- Semiconductor losses coupled to Cauer Thermal Model estimating device temperature
- Healthy operation established as the reference thermal response.



Figure 1.2: Virtual Digital Twin Lab Environment

Degradation Mechanisms [2]

Internal Thermal Degradation

Insulated Gate Bipolar Transistor Electrical Ageing

Thermal Interface

Raman DTS

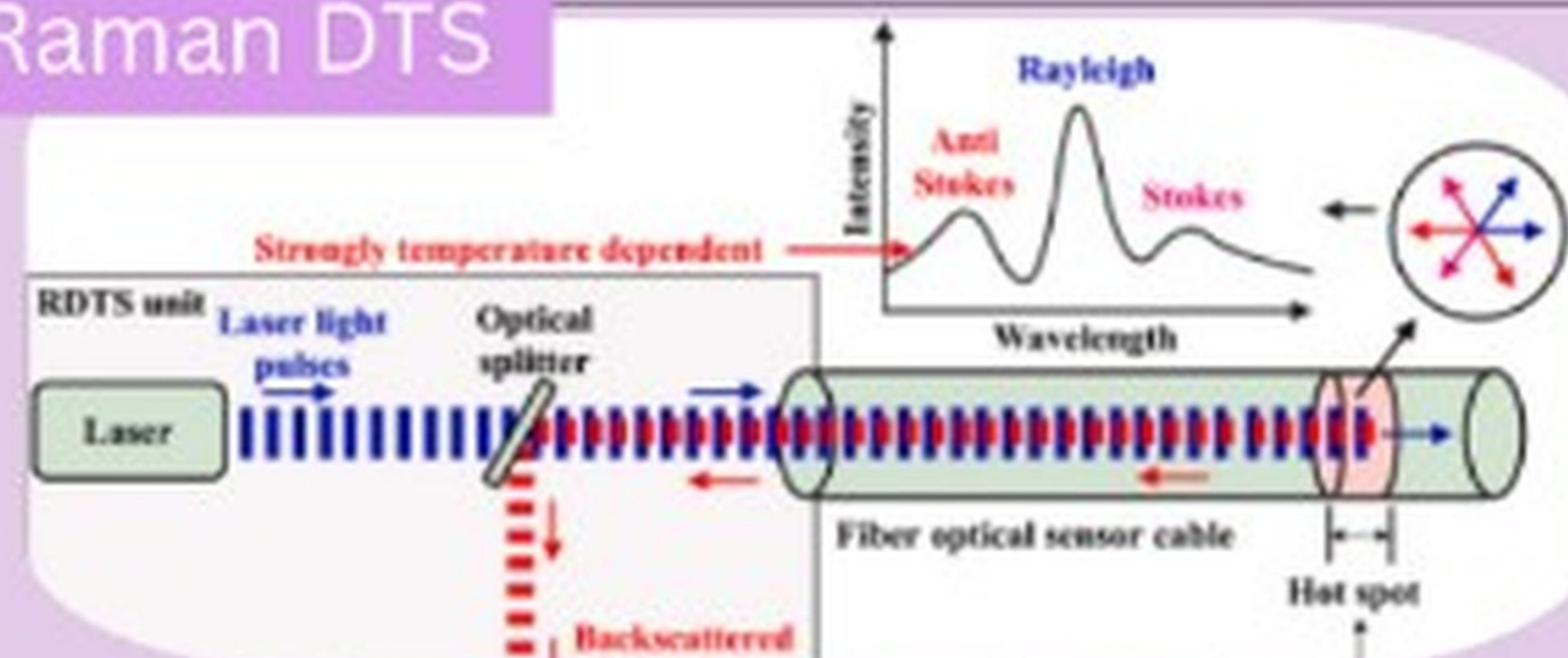
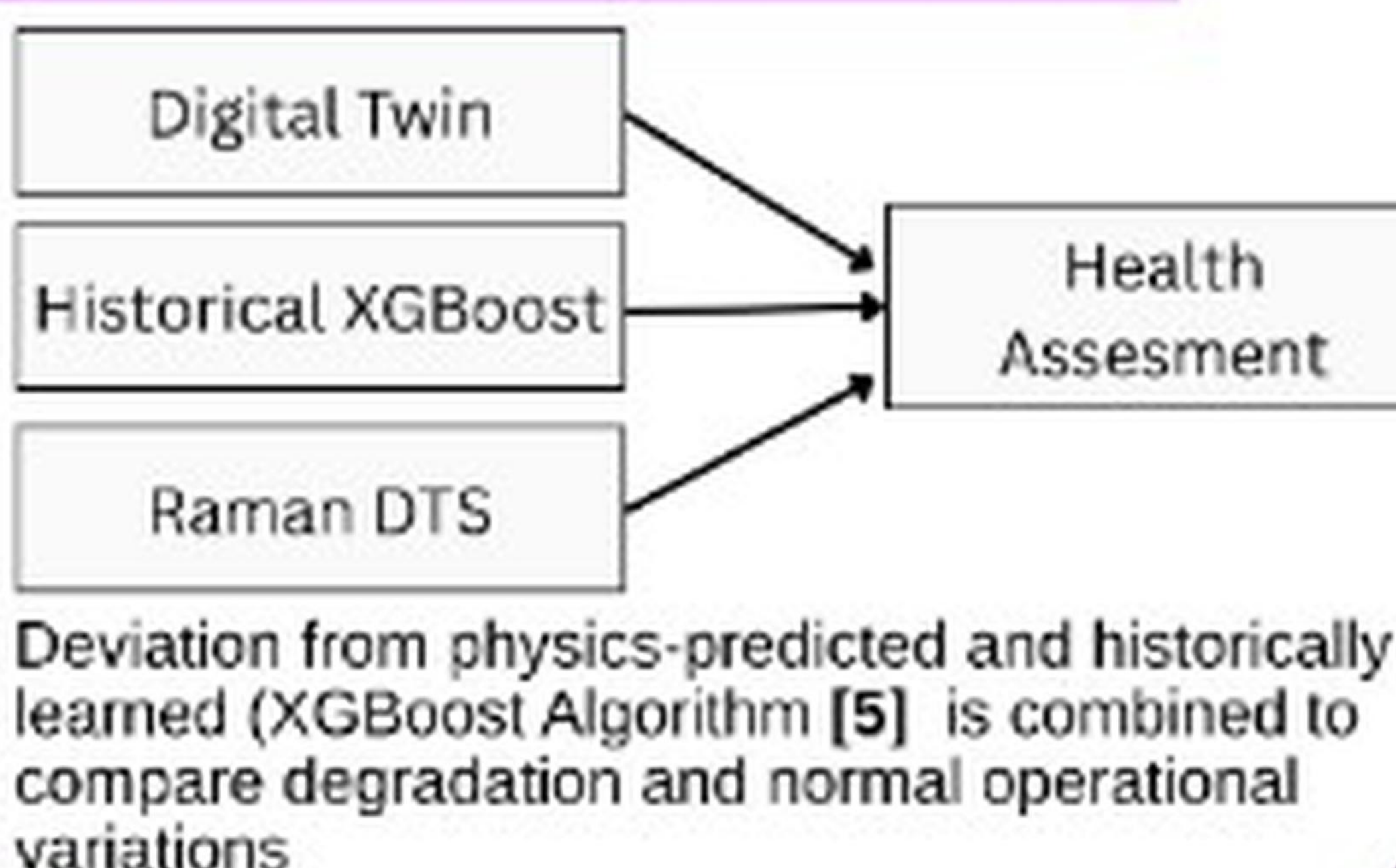


Figure 1.3 Raman Scattering Mechanism

Simulated fibre thermal coupling, Stokes / anti-Stokes Raman Scattering, Optical attenuation [3] and measurement noise [4] reconstruct the spatial inverter temperature profile

Predictive Maintenance Framework



XGBoost



CURRENT RESULTS

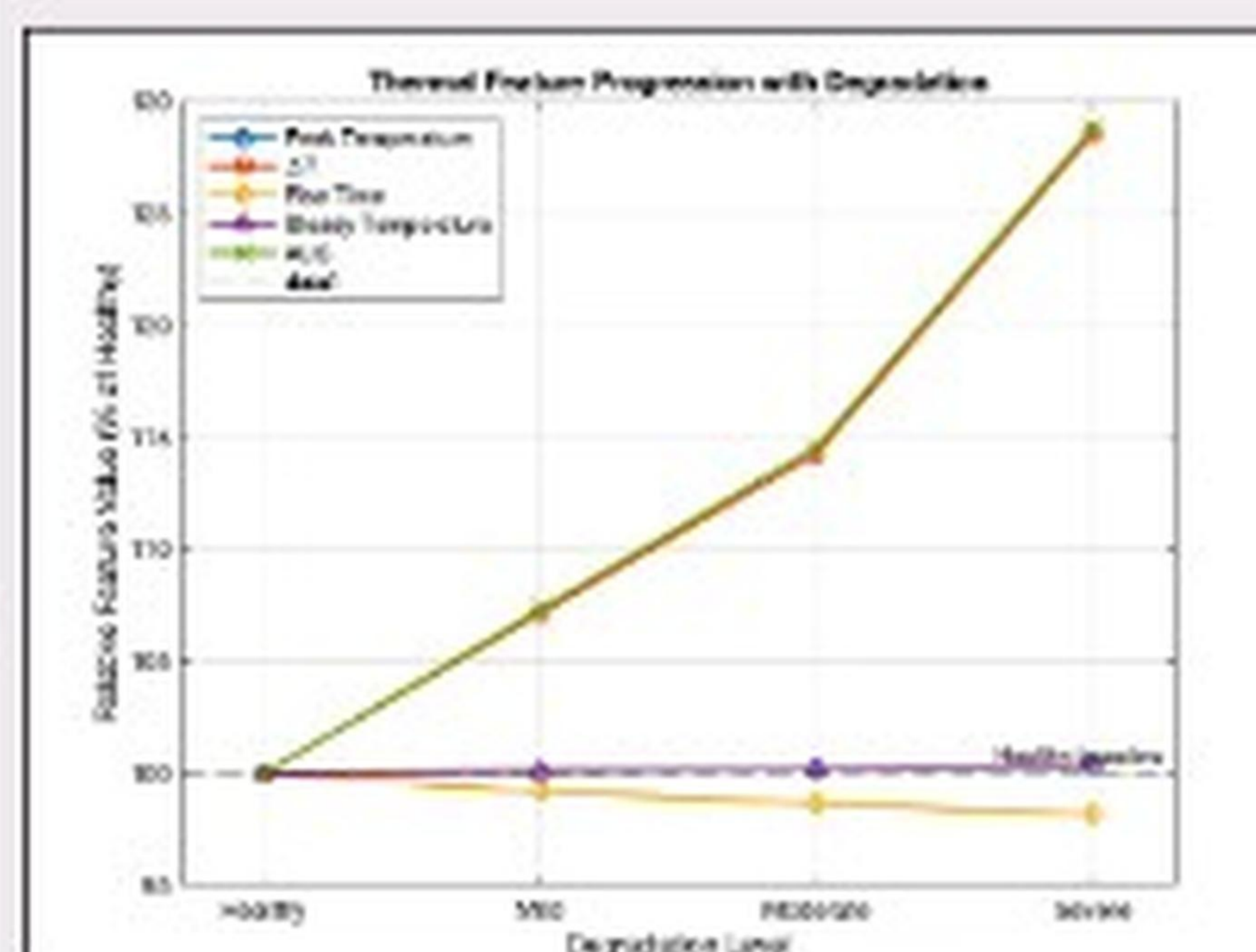


Figure 1.4 Normalised Effect of Degradation Severity on Inverter Thermal Response

$\Delta T \uparrow 28.5\%$
from healthy to severe

$AUC \uparrow 28.8\%$
from healthy to severe

Distinct Severity Trend

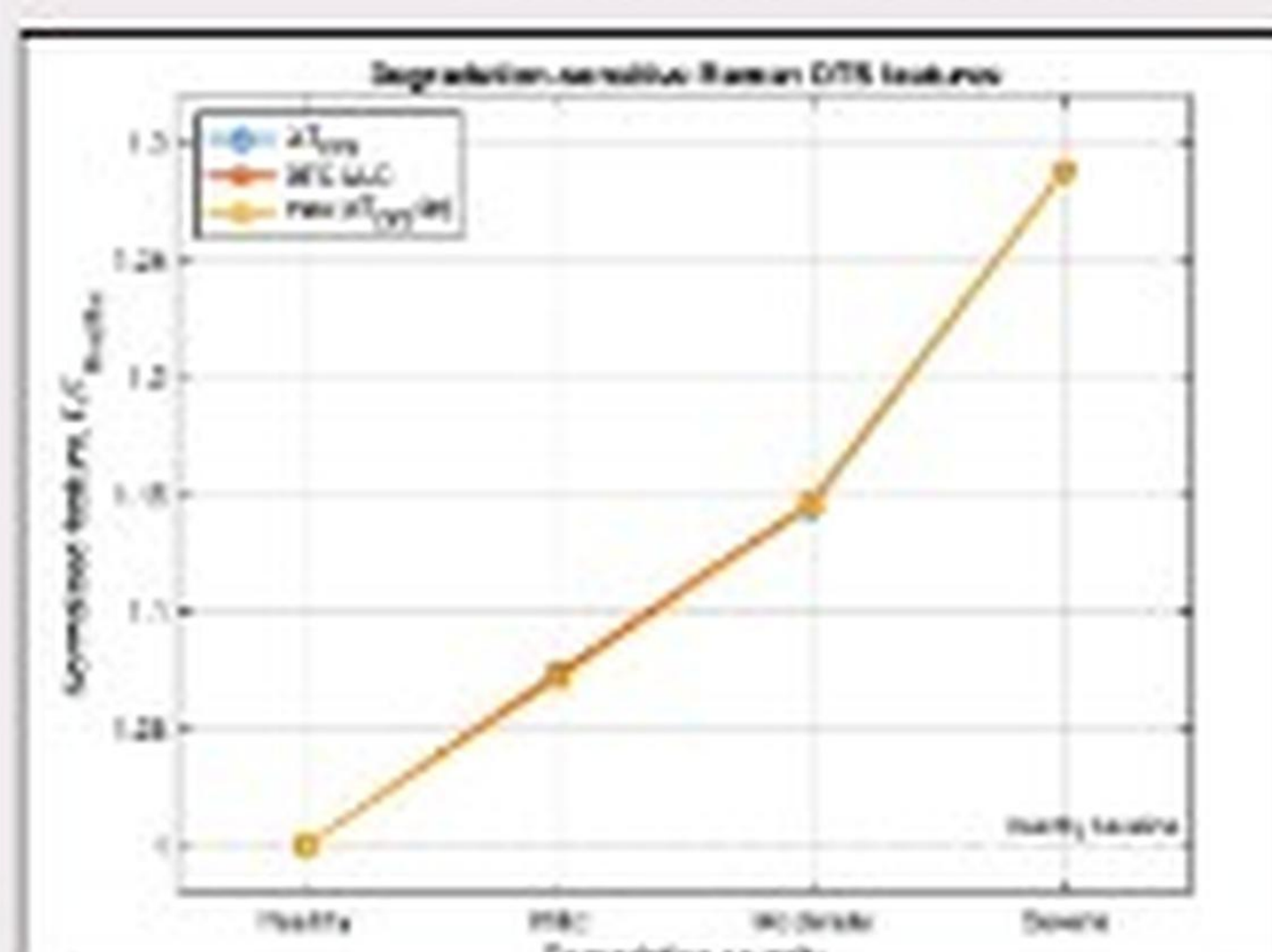


Figure 1.5 Normalisation of Thermal Degradation through Raman DTS Sensing Chain

$\Delta T \uparrow 28.8\%$
from healthy to severe

$AUC \uparrow 28.9\%$
from healthy to severe

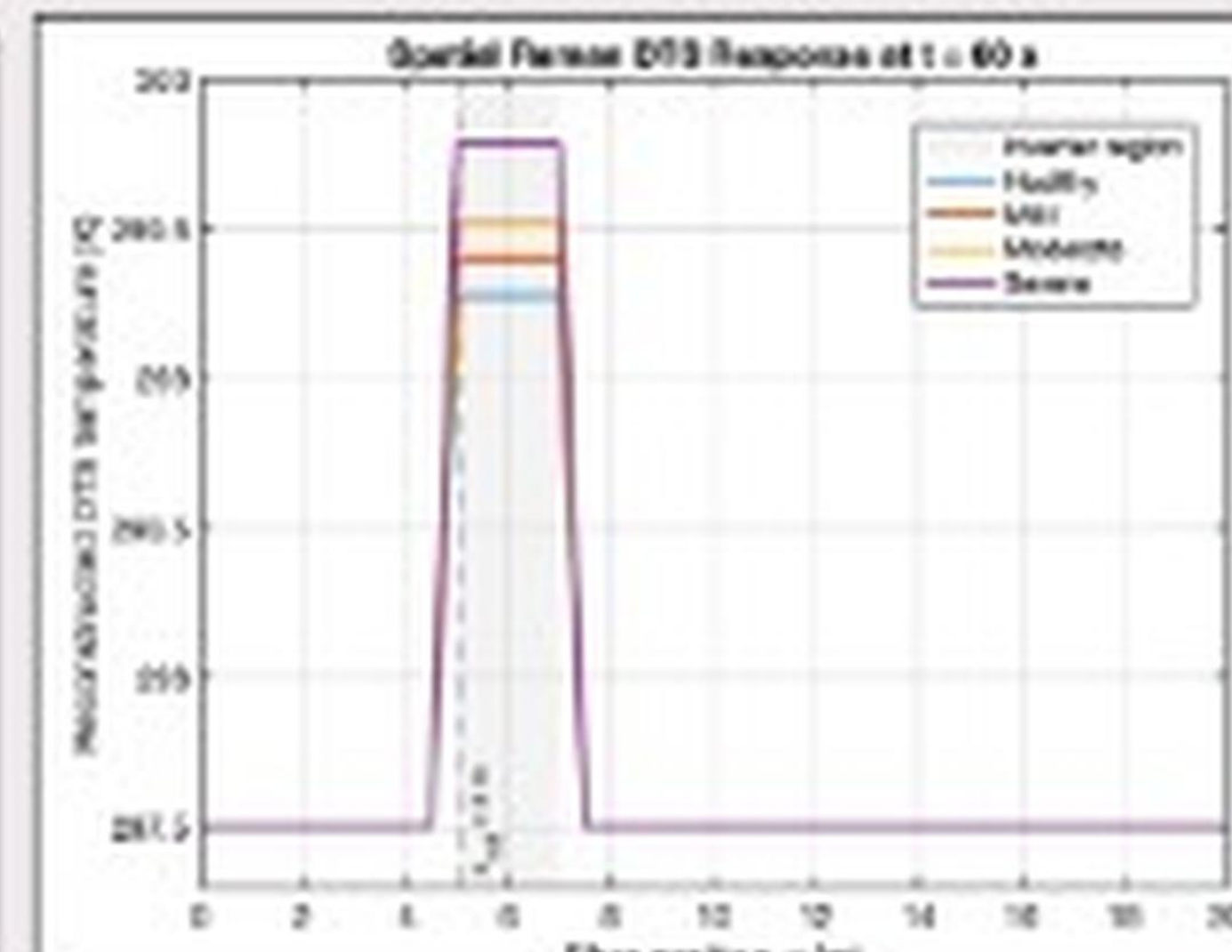


Figure 1.6 Propagation of Thermal Degradation through Raman DTS Sensing Chain

x_{hot}
between 5 to 7 m

Distinct Severity Trend

Next steps are to simulate with longer distances to see the effect of attenuation, add noise and implement XGBoost Algorithm

References

- [1] D. C. Jordan, S. Marion, C. Delme, T. Barnes, and M. Bolinger, "PV field reliability status—Analysis of 100 000 solar systems," *Progress in Photovoltaics: Research and Applications*, vol. 28, no. 8, pp. 739–754, 2020, doi: 10.1002/pip.3262.
- [2] A. Kuprat, K. Dabbas, J. Schaumburg, M. Userra, and M. Langwasser, "Thermal digital twin of power electronics modules for online thermal parameter identification," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 12, no. 1, pp. 1020–1029, Feb. 2024, doi: 10.1109/JESTPE.2023.3328219.
- [3] X. Bao and L. Chen, "Recent Progress in Distributed Fiber Optic Sensors," *Sensors*, vol. 12, no. 7, pp. 8601–8639, Jul. 2012, doi: 10.3390/s120708601.
- [4] S. Jiang et al., "SNR enhancement for Raman distributed temperature sensors using intrinsic modal functions with improved adaptive wavelet threshold denoising," *Optics and Lasers in Engineering*, vol. 189, p. 108940, Jun. 2025, doi: 10.1016/j.optlaseng.2025.108940.
- [5] Trend-Based Predictive Maintenance and Fault Detection Analytics for Photovoltaic Power Plants. Accessed: Jul. 16, 2025. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1002/tpa.202500473>